Standard Data Products

APPN Aerial SOP — Locked revision 1.00

Standard Data Products

This page provides an overview of the **standard data products** generated by APPN UAV processing pipelines, including typical spatial resolution, file formats, and software compatibility. By default, products are delivered in **open or widely supported formats** that can be accessed using **freely available tools**.

For the pipelines that generate these products, see Processing Pipelines.

For additional information about all GOBI and CALVIS products see the Gryfn Documentation

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Guiding principles

- Products are generated using standardised, documented pipelines to ensure consistency across platforms and sites.
- Output formats prioritise accessibility using freely available and open-source software.
- Spatial resolution and product definitions are explicit and traceable.
- Products are intended to be suitable for downstream analysis, comparison, and reuse.

Standard LiDAR-derived products

Product	Description	Typical resolution	Typical file size *(indicative)*	Format	Software compatibility
LiDAR Digital Surface Model (DSM)	Surface model representing canopy and above-ground features	8 cm	Derived from LiDAR point cloud (see below)	GeoTIFF (`.tif`)	QGIS, ArcGIS, GDAL
LiDAR Digital Terrain Model (DTM)	Ground surface model with vegetation removed	1 m	Derived from LiDAR point cloud (see below)	GeoTIFF (`.tif`)	QGIS, ArcGIS, GDAL
Combined LiDAR Point Cloud	Aggregated and filtered point cloud	Native point spacing	~110–175 MB per minute of flight (first & last files may be smaller)	LAS (`.las`)	QGIS, CloudCompare, PDAL

Typical LiDAR file sizes are based on raw LiDAR logging rates documented in the CALViS and GOBI SOPs; DSM and DTM sizes scale with survey area and output resolution.

Standard hyperspectral products

Product	Description	Typical resolution	Typical file size *(indicative)*	Format	Software compatibility
VNIR Orthomosaic	Radiometrically calibrated VNIR hyperspectral mosaic	~4 cm (GSD-based)	Several GB per flight (even for short missions)	ENVI (`.bin` + `.hdr`)	QGIS, ENVI, SNAP
SWIR Orthomosaic	Radiometrically calibrated SWIR hyperspectral mosaic	~4 cm (GSD-based)	Several GB per flight (even for short missions)	ENVI (`.bin` + `.hdr`)	QGIS, ENVI, SNAP

Hyperspectral data volumes scale strongly with mission duration, frame period, and number of spectral bands. SOPs note that even short flights routinely generate multi-GB datasets.

Standard RGB products

Product	Description	Typical resolution	Typical file size *(indicative)*	Format	Software compatibility
RGB Orthomosaic	High-resolution RGB orthorectified mosaic	~0.6 cm (fixed)	~30–70 MB per image (image every ~2 s by default)	GeoTIFF (`.tif`)	QGIS, ArcGIS, GDAL